An Exact Cooperation Formula for Introspection Dynamics in the Heterogeneous Public Goods Game

Harry Foster[1], Vince Knight[1], and Sebastian Krapohl[2]

[1]School of Mathematics, Cardiff University
[2]Faculty of Social and Behavioural Sciences, University of Amsterdam

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Abstract

We study introspection dynamics on the heterogeneous public goods game, where N players may differ in their contributions α_i, public goods multipliers r_i, and choice intensities β_i. We prove that the payoff difference a player evaluates when considering a strategy switch is independent of all other players' current actions, a consequence of the linear payoff structure. This state-independence implies that the introspection Markov chain decomposes as a random-scan product of N independent two-state chains, one per player. The stationary distribution is therefore a product measure, and the long-run cooperation probability admits the exact closed form

$$p_C^{\text{intro}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1-2\mu_i}{1+e^{\beta_i\alpha_i(1-r_i/N)}}+\mu_i\right],$$

with no asymptotic approximation. Several structural consequences follow immediately: a player-specific cooperation threshold at $r_i = N$, neutrality under zero choice intensity, and the sign of each player's sensitivity to their own parameters.

1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed equally, creating a tension between individual incentive and collective benefit [5]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [4].

A natural learning rule for such settings is *introspection dynamics*, introduced in [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played the alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

In [2] the authors extended introspection dynamics to multiplayer asymmetric games and derived a formula for the stationary distribution that can be evaluated numerically for any game. In their general framework the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N; no closed form is available in general.

In this paper we show that the specific payoff structure of the heterogeneous public goods game admits an exact closed-form analysis. The key observation is that the payoff difference $\Delta\pi_i$ a player evaluates depends only on their own current action and individual parameters, not on the actions of others. This is a consequence of the *linearity* of the public goods payoff in contributions. As a result, the introspection chain decomposes exactly as a product of N independent two-state chains (one per player), the stationary distribution is a product measure, and the cooperation probability p_C^{intro} admits an explicit closed form that scales as $O(N)$ rather than $O(2^N)$.

The paper is organised as follows. Section 2 introduces the model. Section 3 establishes the payoff identity. Section 4 proves the product decomposition and derives the exact cooperation formula. Section 5 records several structural corollaries. Section 6 concludes.

2 Setup and notation

Consider a well-mixed population of N ordered players with contributions $\alpha_1, \dots, \alpha_N > 0$, player-specific public goods multipliers $r_1, \dots, r_N > 1$, player-specific choice intensities $\beta_1, \dots, \beta_N > 0$, and two actions $\{C, D\}$ coded as $\{1, 0\}$. The state space is $S = \{0,1\}^N$ with $|S| = 2^N$.

Payoff. The payoff to player i in state $\mathbf{a} = (a_1, \dots, a_N)$ is

$$\pi_i(\mathbf{a}) = \frac{r_i}{N} \sum_{j=1}^{N} \alpha_j a_j - \alpha_i a_i. \tag{1}$$

We write $C(\mathbf{a}) = \sum_j \alpha_j a_j$ for the total contribution.

To give the definition of introspection dynamics we define the following two quantities.

Fermi function. Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$. Since $\beta_i > 0$, the exponential $e^{\beta_i x}$ is strictly increasing in x, so ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$ and $\phi_i(x) + \phi_i(-x) = 1$.

Mutation. Each player i has a mutation rate $\mu_i \in (0, \frac{1}{2})$, ensuring ergodicity of the Markov chain. This is the probability with which they mutate to the other action regardless of the intended choice of the dynamic.

Definition 1 (Cooperation probability)**.** *Given an ergodic chain on S with stationary distribution π, the* cooperation probability *is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N} \sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

Introspection dynamics. At each step, one player i is selected uniformly at random. Player i considers switching from action a_i to the alternative $\bar{a}_i = 1 - a_i$. Writing $\mathbf{a}_{i \to \bar{a}_i}$ for the state obtained by flipping only coordinate i, the transition probability for the switch is

$$T^{\text{intro}}_{\mathbf{a}, \mathbf{a}_{i \to \bar{a}_i}} = \frac{1}{N} \big[\phi_i(\Delta\pi_i)(1 - 2\mu_i) + \mu_i \big], \tag{2}$$

where $\Delta\pi_i := \pi_i(\mathbf{a}) - \pi_i(\mathbf{a}_{i \to \bar{a}_i})$ is the payoff gain from keeping action a_i. Since ϕ_i is strictly decreasing, a larger payoff gain from the current action yields a smaller switching probability.

3 The payoff identity

Lemma 2 (State-independence of payoff differences)**.** *For the payoff* (1)*, the difference player i evaluates when considering a switch from a_i to $\bar{a}_i$ is*

$$\Delta\pi_i = (2a_i - 1)\, \alpha_i \left(\frac{r_i}{N} - 1 \right). \tag{3}$$

This depends only on a_i, α_i, r_i, and N; in particular, it is independent of all other players' actions $\mathbf{a}_{-i}$.

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own contribution changes, so the total contribution becomes $C(\mathbf{a}_{i \to \bar{a}_i}) = C(\mathbf{a}) + \alpha_i(1 - 2a_i)$. Expanding both payoffs:

$$\begin{aligned}
\pi_i(\mathbf{a}) &= \frac{r_i}{N}\, C(\mathbf{a}) - \alpha_i a_i, \\
\pi_i(\mathbf{a}_{i \to \bar{a}_i}) &= \frac{r_i}{N}\big[C(\mathbf{a}) + \alpha_i(1 - 2a_i)\big] - \alpha_i(1 - a_i).
\end{aligned}$$

Subtracting:

$$\begin{aligned}
\Delta\pi_i &= -\frac{r_i}{N}\alpha_i(1 - 2a_i) + \alpha_i(1 - 2a_i) \\
&= \alpha_i(1 - 2a_i)\Big(1 - \frac{r_i}{N}\Big) \\
&= \alpha_i(2a_i - 1)\Big(\frac{r_i}{N} - 1\Big).
\end{aligned}$$

The term $C(\mathbf{a})$ cancels entirely, so $\Delta\pi_i$ is independent of $\mathbf{a}_{-i}$.

□

Remark 3. *The cancellation of $C(\mathbf{a})$ relies on the payoff* (1) *being* linear *in contributions. The heterogeneity of r_i does not affect this cancellation: each player's comparison depends only on their own parameters.*

4 Individual cooperation probabilities

By Lemma 2, player i's update depends only on their own action a_i, not on $\mathbf{a}_{-i}$. We use this to compute each player's long-run cooperation probability directly from their two-state update rule.

Theorem 4 (Individual cooperation probabilities)**.** *Under introspection dynamics with player-specific β_i and r_i, uniform mutation $\mu_i \in (0, \frac{1}{2})$, and payoff* (1)*, the long-run probability that player i cooperates is*

$$p_i = \phi_i\big(\alpha_i(1 - r_i/N)\big)(1 - 2\mu_i) + \mu_i. \tag{4}$$

Proof. By Lemma 2, $\Delta\pi_i$ depends only on a_i. When player i is selected, the probability their next action is C equals p_i regardless of their current action: if $a_i = D$, the switch probability is $\phi_i(\alpha_i(1-r_i/N))(1-2\mu_i)+\mu_i = p_i$; if $a_i = C$, the probability of remaining at C is $1-[\phi_i(\alpha_i(r_i/N-1))(1-2\mu_i)+\mu_i] = p_i$, using $\phi_i(x)+\phi_i(-x) = 1$. Since each selection of player i places them in state C with probability p_i independently of their current state, the long-run cooperation probability of player i is p_i.

□

Corollary 5 (Exact cooperation probability)**.** *The long-run cooperation probability is*

$$\boxed{p_C^{\text{intro}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1 - 2\mu_i}{1 + e^{\beta_i\alpha_i(1 - r_i/N)}} + \mu_i\right].} \tag{5}$$

Proof. By Theorem 4:

$$p_C^{\mathrm{intro}} = \frac{1}{N}\sum_{i=1}^{N} p_i.$$

Substituting $\phi_i(\alpha_i(1 - r_i/N)) = (1 + e^{\beta_i \alpha_i (1-r_i/N)})^{-1}$ into (4) gives (5).

□

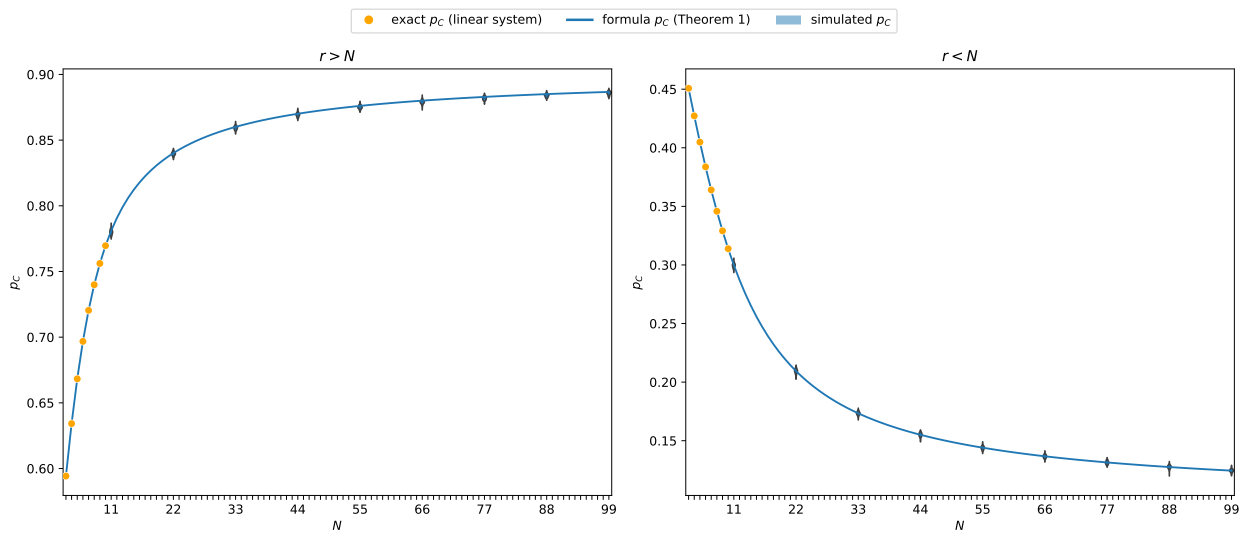

Figure 1: Validation of the exact formula (5) on a heterogeneous group with contributions $\alpha_i = i$ for $i = 0, 1, \ldots, N-1$, choice intensity $\beta = 0.5$, and mutation rate $\mu_i = 0.1$. *Left:* $r = 2N$ (so $r > N$ for all N). *Right:* $r = N/2$ (so $r < N$ for all N). Orange dots: exact p_C obtained by solving the full 2^N linear system (feasible only for small N). Blue curve: the closed-form formula (5) (Theorem 4). Blue bands: cooperation probability estimated from 10^5 simulation steps across 99 independent runs; the band width reflects run-to-run variability. The exact and the estimated values of p_C are obtained using [3]. The formula matches both the exact values and the simulations exactly.

5 Structural consequences

The closed form (5) allows immediate structural conclusions about how cooperation depends on game parameters.

Corollary 6 (Player-specific cooperation thresholds)**.** *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if $r_i > N$. A sufficient condition for $p_C^{\mathrm{intro}} > \frac{1}{2}$ is $r_i > N$ for all i.*

Proof. Since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$:

$$p_i > \tfrac{1}{2} \iff \phi_i\big(\alpha_i(1 - r_i/N)\big) > \tfrac{1}{2} \iff \alpha_i(1 - r_i/N) < 0 \iff r_i > N.$$

The average $p_C^{\mathrm{intro}} = \frac{1}{N}\sum_i p_i$ exceeds $\frac{1}{2}$ whenever every summand does.

□

Corollary 7 (Neutral drift)**.** *If $\beta_i = 0$ for all i, then $p_C^{\mathrm{intro}} = \frac{1}{2}$ regardless of all other parameters.*

Proof. When $\beta_i = 0$, the Fermi function satisfies $\phi_i(x) = \frac{1}{2}$ for all x. Substituting into (5):

$$p_C^{\text{intro}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1-2\mu_i}{2} + \mu_i\right] = \frac{1-2\mu_i}{2} + \mu_i = \frac{1}{2}. \qquad \square$$

Corollary 8 (Role of heterogeneity)**.** *For player i with $r_i \neq N$:*

(i) $\dfrac{\partial p_i}{\partial \alpha_i}$ *is positive when $r_i > N$ and negative when $r_i < N$.*

(ii) $\dfrac{\partial p_i}{\partial \beta_i}$ *is positive when $r_i > N$ and negative when $r_i < N$.*

Proof. Write $p_i = (1-2\mu_i)\phi_i(\alpha_i(1-r_i/N)) + \mu_i$.

Part (i). Differentiating with respect to α_i:

$$\frac{\partial p_i}{\partial \alpha_i} = (1-2\mu_i)(1-r_i/N)\,\phi_i'(\alpha_i(1-r_i/N)).$$

Since $\phi_i'(x) < 0$ for all x and $1-2\mu_i > 0$, the sign equals $-\mathrm{sgn}(1-r_i/N) = \mathrm{sgn}(r_i/N-1)$, which is positive if and only if$r_i > N$.

Part (ii). Write $g(t) = (1+e^t)^{-1}$, so $\phi_i(x) = g(\beta_i x)$. Differentiating with respect to β_i:

$$\frac{\partial p_i}{\partial \beta_i} = (1-2\mu_i)\,\alpha_i(1-r_i/N)\,g'(\beta_i\alpha_i(1-r_i/N)).$$

Since $g' < 0$ and $\alpha_i > 0$, the sign equals $-\mathrm{sgn}(1-r_i/N) = \mathrm{sgn}(r_i/N-1)$, which is positive if and only if $r_i > N$. $\square$

Remark 9. *Corollaries 6–8 show that each player's long-run cooperation is fully determined by their individual triple (α_i, β_i, r_i) and the group size N. Players with $r_i > N$ cooperate more than half the time, and increasing α_i or β_i raises their cooperation probability further; players with $r_i < N$ show the opposite sensitivity. The parameters of other players have no effect on player i's marginal cooperation probability.*

6 Conclusion

We have proved that introspection dynamics on the heterogeneous public goods game is exactly solvable. The linear payoff structure implies that each player's payoff comparison is independent of all other players' actions (Lemma 2), so when player i is selected their next action is C with probability p_i regardless of current state (Theorem 4). The long-run cooperation probability therefore has a closed form with N terms rather than 2^N (Corollary 5).

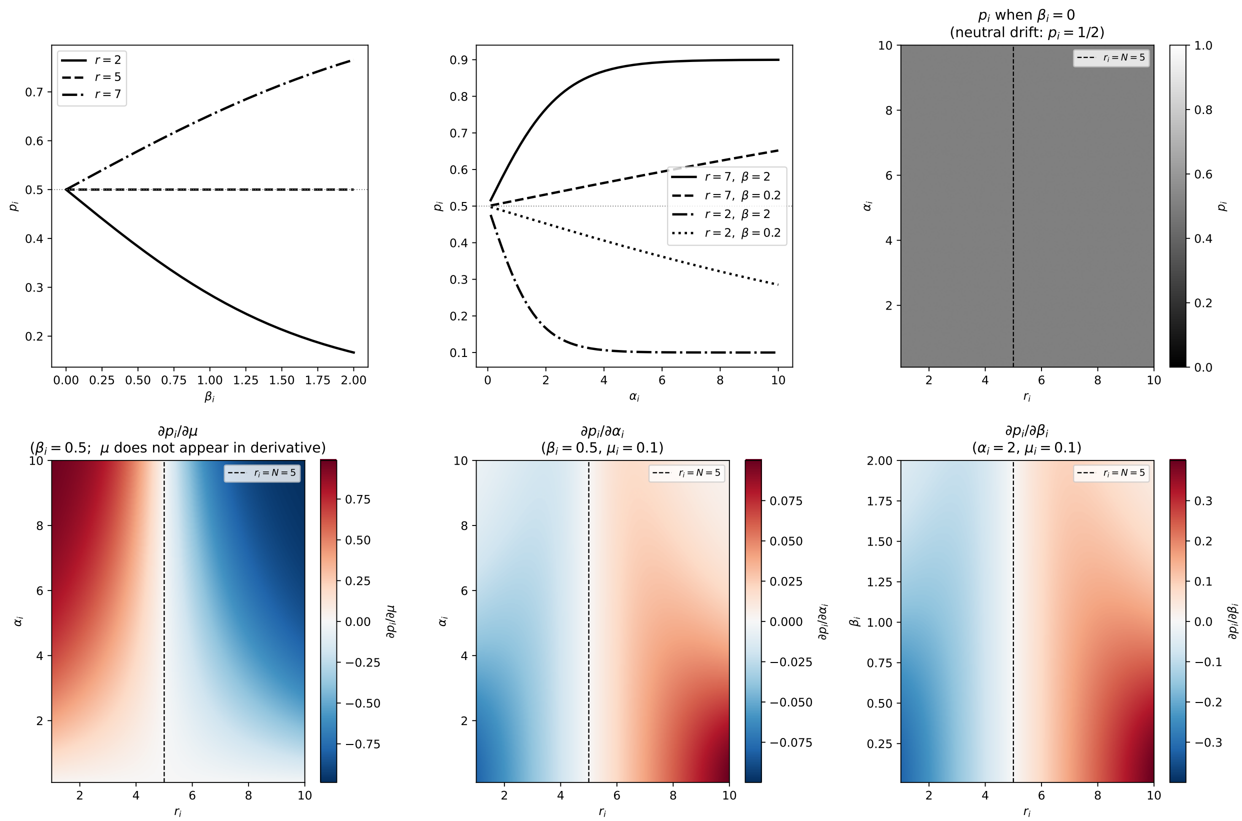

Figure 2: Structural properties of p_i for a single player with $N = 5$, $\mu_i = 0.1$. *Top left:* p_i as a function of β_i ($\alpha_i = 2$) for $r < N$, $r = N$, and $r > N$; all curves pass through $p_i = \frac{1}{2}$ at $\beta_i = 0$ (Corollary 7). *Top centre:* p_i as a function of α_i for two values of r and β_i; curves above (below) $\frac{1}{2}$ correspond to $r > N$ ($r < N$). *Top right:* $p_i(\alpha_i, r_i)$ at $\beta_i = 0$, confirming $p_i = \frac{1}{2}$ everywhere (Corollary 7); the dashed line marks $r_i = N$. *Bottom row:* Heatmaps of the partial derivatives $\partial p_i/\partial \mu_i$, $\partial p_i/\partial \alpha_i$, and $\partial p_i/\partial \beta_i$ over the (r_i, α_i) or (r_i, β_i) plane ($\beta_i = 0.5$ or $\alpha_i = 2$ as fixed parameter). Red indicates a positive derivative; blue indicates negative. Each derivative changes sign exactly at $r_i = N$ (dashed line), consistent with Corollary 8.

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